Chap 5
Integrals

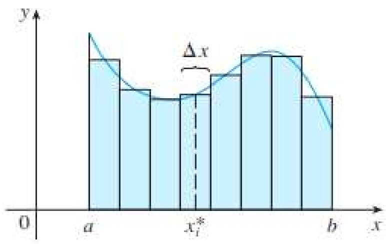
§2 The Definite Integrals

* <u>Definition of a Definite Integrals</u>

f : a function defined for $a \le x \le b$

$\triangle x = \dfrac{b-a}{n}$ and $x_i = a + i\triangle x$ $(i = 0, 1, 2, \cdots, n)$

$x_i^* \in [x_{i-1},\ x_i]$: sample points $(i = 1, 2, \cdots, n)$



$\Rightarrow$ The **Definite Integral of** f **from** a **to** b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\triangle x$$

provided that this limit exists and gives the same value for all possible choices of sample points.

If it does exist, we say that f is **integrable** on $[a,\ b]$.

<u>Note</u>

1. $\displaystyle\int$: integral sign

   $f(x)$: integrand

   $a,\ b$: limits of integration

   (a : lower limit, b : upper limit)

2. $\displaystyle\int_a^b f(x)\,dx = \int_a^b f(t)\,dt = \int_a^b f(r)\,dr = \cdots$

3. $\displaystyle\sum_{i=1}^{n} f(x_i^*)\triangle x$: Riemann sum

Theorem 3 If f is continuous on $[a, b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a, b]$.

Theorem 4 If f is integrable on $[a, b]$, then

$$\begin{aligned}\int_a^b f(x)\,dx &= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\,\triangle x \\ &= \lim_{n\to\infty}\sum_{i=0}^{n-1} f(x_i)\,\triangle x\end{aligned}$$

Exam 1 Express

$$\lim_{n\to\infty}\sum_{i=1}^{n}(x_i^3 + x_i\sin x_i)\,\triangle x$$

as an integral on the interval $[0, \pi]$.

(sol) Let $f(x) = x^3 + x\sin x$, which is continuous on $[0, \pi]$.

$$\begin{aligned}\Rightarrow\ \lim_{n\to\infty}\sum_{i=1}^{n}(x_i^3 + x_i\sin x_i)\,\triangle x &= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\,\triangle x \\ &= \int_0^{\pi} f(x)\,dx \\ &= \int_0^{\pi}(x^3 + x\sin x)\,dx\end{aligned}$$

<u>Exam 2</u> (a) Evaluate the Riemann sum for $f(x)=x^3-6x$, taking the sample points to be right endpoints and $a=0$, $b=3$, and $n=6$.

(b) Evaluate $\int_0^3 (x^3-6x)dx$.

(sol) (a) The subinterval width is

$$\triangle x=\frac{3-0}{6}=\frac{1}{2}$$

and the right endpoints are

$$x_1=0.5,\quad x_2=1,\quad x_3=1.5,\quad x_4=2,\quad x_5=2.5,\quad x_6=3.$$

So the Riemann sum is

$$\begin{aligned} R_6 &= \sum_{i=1}^{6} f(x_i)\triangle x \\ &= f(0.5)\cdot 0.5+f(1)\cdot 0.5+\cdots+f(3)\cdot 0.5 \\ &= -3.9375 \end{aligned}$$

(b) The subinterval width is

$$\triangle x=\frac{3-0}{n}=\frac{3}{n}$$

and the right endpoints are

$$x_i=0+i\triangle x=\frac{3}{n}i \qquad (i=1,2,\cdots,n)$$

Let $f(x)=x^3-6x$.

$$\begin{aligned} \Rightarrow \int_0^3 (x^3-6x)dx &= \int_0^3 f(x)dx \\ &= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\triangle x \\ &= \lim_{n\to\infty}\sum_{i=1}^{n} (x_i^3-6x_i)\triangle x \\ &= \lim_{n\to\infty}\sum_{i=1}^{n}\left\{\left(\frac{3i}{n}\right)^3-6\left(\frac{3i}{n}\right)\right\}\frac{3}{n} \\ &= \lim_{n\to\infty}\left[\frac{3^4}{n^4}\sum_{i=1}^{n} i^3-\frac{54}{n^2}\sum_{i=1}^{n} i\right] \\ &= \lim_{n\to\infty}\left[\frac{3^4}{n^4}\left\{\frac{n(n+1)}{2}\right\}^2-\frac{54}{n^2}\,\frac{n(n+1)}{2}\right] \\ &= \frac{81}{4}-\frac{54}{2} \\ &= -\frac{27}{4}=-6.75 \end{aligned}$$

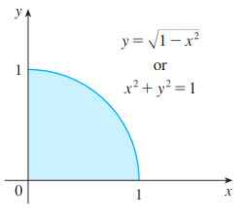
<u>Exam 3</u> Evaluate the following integrals by interpreting each in terms of areas.

(a) $\int_0^1 \sqrt{1-x^2}\, dx$ (b) $\int_0^3 (x-1)\, dx$

(sol)

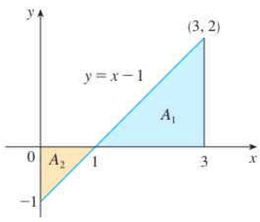
(a) Let $y = \sqrt{1-x^2}$.

$\Rightarrow\ x^2+y^2=1, \quad y \ge 0$



$\therefore\ \int_0^1 \sqrt{1-x^2}\, dx = \frac{\pi}{4}$

(b) Let $y = x-1$



$$\therefore\ \int_0^3 (x-1)\, dx = -A_2 + A_1$$
$$= -\frac{1}{2}+2$$
$$= \frac{3}{2}$$

$*$ <u>The Midpoint Rule</u>

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} f(\overline{x_i})\,\triangle x$$

where $\triangle x = \dfrac{b-a}{n}$ and $\overline{x_i} = \dfrac{x_{i-1}+x_i}{2}$ = the midpoint of $[x_{i-1},\ x_i]$.

<u>Exam 4</u> Use the Midpoint Rule with $n=5$ to approximate $\displaystyle\int_1^2 \frac{1}{x}\,dx$.

(sol) $\triangle x = \dfrac{2-1}{5} = 0.2$ and

$$x_0 = 1,\ x_1 = 1.2,\ x_2 = 1.4,\ x_3 = 1.6,\ x_4 = 1.8,\ x_5 = 2$$

Then the midpoints are

$$\overline{x_1} = 1.1,\ \overline{x_2} = 1.3,\ \overline{x_3} = 1.5,\ \overline{x_4} = 1.7,\ \overline{x_5} = 1.9$$

$$\begin{aligned}
\therefore\ \int_1^2 \frac{1}{x}\,dx &\approx \sum_{i=1}^{5} f(\overline{x_i})\,\triangle x \\
&= [f(1.1)+f(1.3)+f(1.5)+f(1.7)+f(1.9)]\times 0.2 \\
&= \left[\frac{1}{1.1}+\frac{1}{1.3}+\frac{1}{1.5}+\frac{1}{1.7}+\frac{1}{1.9}\right]\times 0.2 \\
&\approx 0.691908
\end{aligned}$$

where the error is 0.001239 since the true value is about 0.693147.

(cf)

(i) $L_5 = \left[\dfrac{1}{1}+\dfrac{1}{1.2}+\dfrac{1}{1.4}+\dfrac{1}{1.6}+\dfrac{1}{1.8}\right]\times 0.2$ with error ≈ 0.052488

≈ 0.745635

(ii) $R_5 = \left[\dfrac{1}{1.2}+\dfrac{1}{1.4}+\dfrac{1}{1.6}+\dfrac{1}{1.8}+\dfrac{1}{2}\right]\times 0.2$ with error ≈ 0.047512

≈ 0.645635

∗ <u>Properties of the Definite Integral</u>

- $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.
- $\int_a^a f(x)\,dx = 0$.

1. $\int_a^b c\,dx = c(b-a)$. (c : constant)
2. $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$.
3. $\int_a^b [cf(x)]\,dx = c\int_a^b f(x)\,dx$.
4. $\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$.
5. $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$.
6. If $f(x) \ge 0$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \ge 0$.
7. If $f(x) \ge g(x)$ for $a \le x \le b$, then
$$\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx.$$
8. If $m \le f(x) \le M$ for $a \le x \le b$, then
$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

- The properties of 6-8 are called the **Comparison Properties of the integral**.

<u>**Exam 6**</u> If it is known that $\int_0^{10} f(x)\,dx = 17$ and $\int_0^8 f(x)\,dx = 12$, find $\int_8^{10} f(x)\,dx$.

(sol) Since $\int_0^8 f(x)\,dx + \int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx$,
$$\int_8^{10} f(x)\,dx = \int_0^{10} f(x)\,dx - \int_0^8 f(x)\,dx = 17-12 = 5.$$